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Litterature:

H. H. Hamedani, *Probability and Statistics: The Science of Uncertainty*, 3rd ed.,
<https://www.probabilitycourse.com>

Course: Stokastisk modellering og behandling (SWSTS-01)

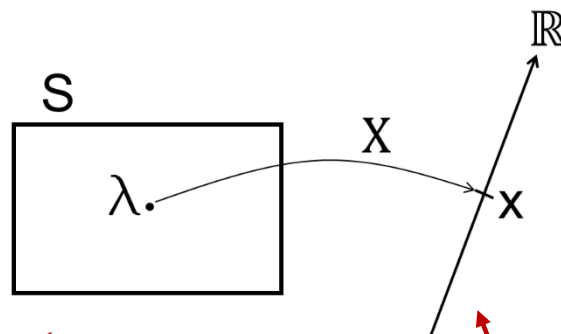
Random variable

A **random variable** is a real-valued function that assigns a number to each outcome of a random experiment.

$$X: S \rightarrow \mathbb{R}$$

Such that $X(\lambda) = x$ is understood as:

“For a particular outcome λ the function value is x ”



Domain / sample space — S (pre-measurement world)

- Outcomes not yet observed.
- “Physical reality” of the experiment.
- Often complicated / abstract.

Examples:

- Coin sequences
- Noise realizations
- Microstates of a system

Range — $\mathbb{R}$ (post-measurement world)

- Numerical value after observation.
- What we record, plot, and model.

Examples:

- Number of heads
- Voltage measurement
- Stock return

Random variable

Example: Toss a coin 5 times

- Sample space:

$$S = \{TTTTT, TTTTH, \dots, HHHHH\}$$

(Total of $2^5 = 32$ outcomes)



Let X be the number of heads in the outcome.

Then $X \in \{0, 1, 2, 3, 4, 5\}$

- For example:

- $X(TTTTT) = 0$
- $X(THTHT) = 2$
- $X(HHHHH) = 5$

Example: Rolling two Dices

Sample space S :

$$S = \{(1, 1), (1, 2), \dots, (6, 6)\}$$

(There are 36 total outcomes.)



Let X be the **sum** of the numbers rolled on two dice.

For example:

$$X(1, 1) = 2, X(2, 5) = 7, X(6, 6) = 12$$

Range R_X of X :

The set of all possible values that the random variable X can take:

$$R_X = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

Random variable

Example: Throwing a Dart at a Target

A dart is thrown at a circular target of radius 3. Assume the dart always hits the target.



Sample space

$$S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 9\}$$

(All possible landing points on the target.)

Let X be the distance from the centre

Define the random variable

$$X(x, y) = \sqrt{x^2 + y^2}.$$

So X measures how far the dart lands from the origin.

For example

- $X(0, 0) = 0$ (bullseye)
- $X(3, 0) = 3$ (edge of target)

Range R_X of X

The set of all possible values that X can take:

$$R_X = [0, 3].$$

Nb. A random variable is **discrete** if it's range is countable.

A random variable is **continuous** if it's range is uncountable.

Probability Mass Function (PMF)

The PMF gives the probability of each possible value of a discrete random variable:

$$f_X(x_k) = P(X = x_k), \quad \text{for each } x_k \in R_X$$

Properties:

$$0 \leq f_X(x) \leq 1, \quad \sum f_X(x_i) = 1$$

Example

In Lecture 3, you learned the **Binomial Formula**:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

This is a **PMF**! It gives the probability of getting exactly k successes (e.g., heads) in n independent trials.

Let X be the number of **heads** in 3 tosses.

This is a **binomial distribution** with parameters:

- $n = 3$ (trials)
- $p = 0.5$ (probability of heads)



$$P(X = k) = \begin{cases} 0.125 & \text{if } k = 0 \\ 0.375 & \text{if } k = 1 \\ 0.375 & \text{if } k = 2 \\ 0.125 & \text{if } k = 3 \\ 0 & \text{otherwise} \end{cases}$$

Probability Mass Function (PMF)

Example: Fair dice

Let X be the “number of eyes” on a die. Individual probabilities of each random variable:

$$P_X(x_k = 0) = \frac{1}{6}$$

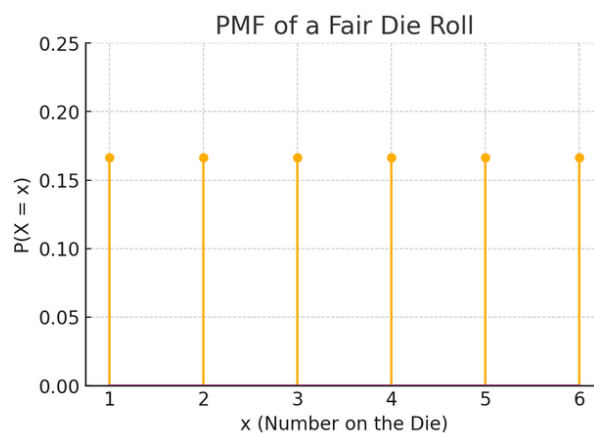
$$P_X(x_k = 1) = \frac{1}{6}$$

$\vdots$

$$P_X(x_k = 6) = \frac{1}{6}$$

Expressed as a **probability mass function** (PMF), and the associated graph:

$$P(X = x) = \begin{cases} \frac{1}{6} & \text{if } x = 1, 2, 3, 4, 5, 6 \\ 0 & \text{otherwise} \end{cases}$$



Probability Mass Function (PMF)

Example: Binomial formula and **two** fair dice

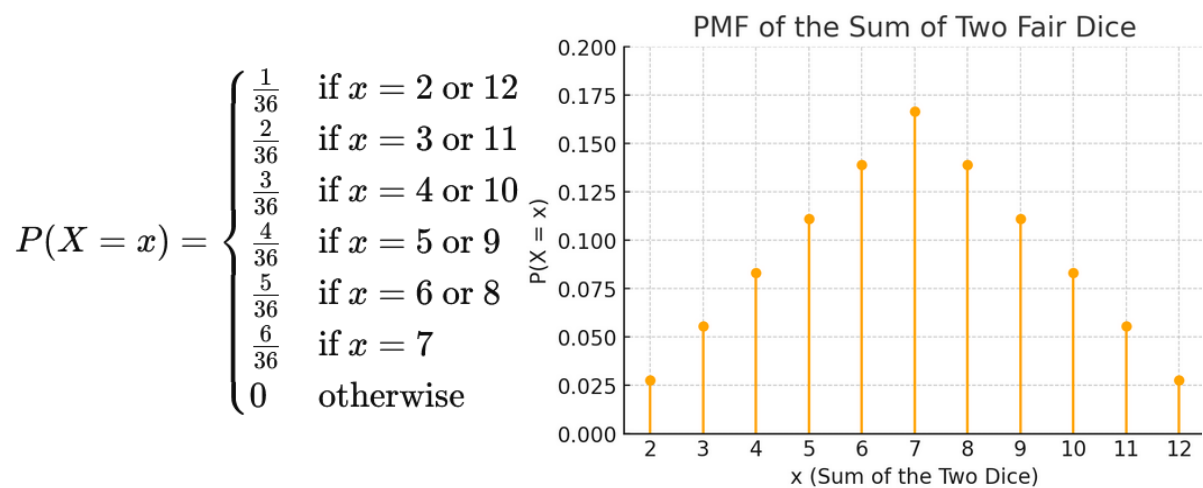
Let X be the “number of eyes” on both dices.

There are $6 \times 6 = 36$ equally likely outcomes when two dice are rolled.

We compute the PMF by **counting** the number of favorable outcomes for each sum:

$$P(X = x) = \frac{\text{Number of ways to get sum } x}{36}$$

We can use this formula to calculate the **probability mass function (PMF)**, and the associated graph:



Quiz

What is the total sum of all probabilities in this valid PMF?

$$P(X = x) = \begin{cases} \frac{1}{36} & \text{if } x = 2 \text{ or } 12 \\ \frac{2}{36} & \text{if } x = 3 \text{ or } 11 \\ \frac{3}{36} & \text{if } x = 4 \text{ or } 10 \\ \frac{4}{36} & \text{if } x = 5 \text{ or } 9 \\ \frac{5}{36} & \text{if } x = 6 \text{ or } 8 \\ \frac{6}{36} & \text{if } x = 7 \\ 0 & \text{otherwise} \end{cases}$$

A: 0.1

B: 0.5

C: 1

D: 1.5

Quiz

One probability is missing in the following PMF. Find its value so that the distribution is valid.

$$P(X = x) = \begin{cases} \frac{1}{10}, & x = 1 \\ \frac{2}{10}, & x = 2 \\ ?, & x = 3 \\ \frac{3}{10}, & x = 4 \\ \frac{1}{10}, & x = 5 \\ 0, & \text{otherwise} \end{cases}$$

A: 2/10

B: 3/10

C: 1/10

D: 0

Special Distributions (PMF)

List of special **discrete distributions** (or Probability Mass Functions **PMF**) for this course:

1. Bernoulli Distribution

- Models a random experiment with **two outcomes**: Success (1) or Failure (0).
- Example: Pass/fail test, coin toss, gender of a baby.
- PMF:

$$P_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{where } 0 < p < 1$$

- Notation: $X \sim \text{Bernoulli}(p)$

2. Binomial Distribution

- Models the total number of **successes** in n independent Bernoulli trials, each with success probability p .
- Example: Number of heads in 10 coin tosses, number of correct answers on a multiple-choice test.
- PMF:

$$P_X(k) = \begin{cases} \binom{n}{k} p^k (1 - p)^{n-k} & \text{if } k = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases} \quad \text{where } 0 < p < 1$$

- Notation: $X \sim \text{Binomial}(n, p)$

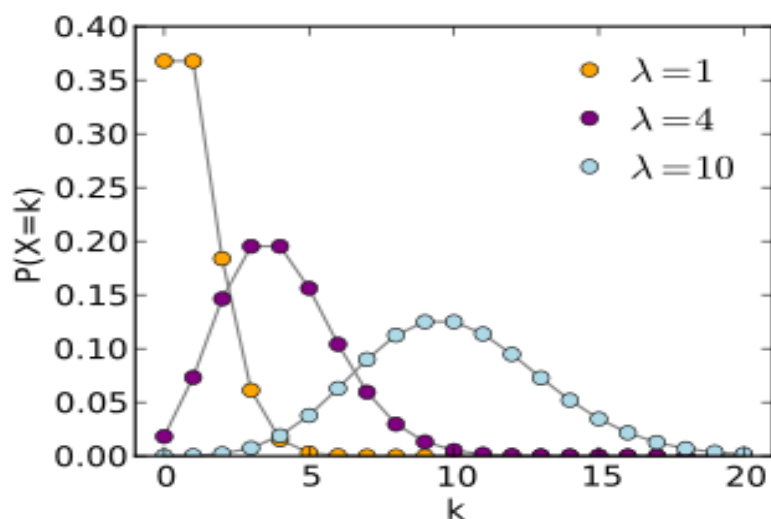
Special Distributions (PMF)

3. Poisson Distribution

- Models number of events in a fixed time or space interval.
- Parameters:
 - $\lambda > 0$: average number of events in the interval
 - $k \in \{0, 1, 2, \dots\}$: actual observed number of events
- PMF:

$$P_X(k) = \begin{cases} \frac{e^{-\lambda} \lambda^k}{k!} & \text{for } k = 0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

- Notation: $X \sim \text{Poisson}(\lambda)$



Special Distributions (PMF)**Example:** Poisson Distribution in Practice

An average of **0.2 emails per minute** arrives on a weekday. We model this with a **Poisson distribution**.

Questions:

1. What is the probability of receiving **no emails in 5 minutes**?
2. What is the probability of receiving **more than 3 emails in 10 minutes**?

Solution**1. Probability of no emails in 5 minutes**

Let X be the number of emails in a 5-minute interval.

Then $X \sim \text{Poisson}(\lambda = 5 \cdot 0.2 = 1)$.

$$P(X = 0) = \frac{e^{-1} \cdot 1^0}{0!} = \frac{e^{-1}}{1} = \frac{1}{e} \approx 0.3679$$

Special Distributions (PMF)

2. Probability of more than 3 emails in 10 minutes

Let Y be the number of emails in a 10-minute interval.

Then $Y \sim \text{Poisson}(\lambda = 10 \cdot 0.2 = 2)$.

We want $P(Y > 3) = 1 - P(Y \leq 3)$:

$$\begin{aligned}
 P(Y > 3) &= 1 - [P_Y(0) + P_Y(1) + P_Y(2) + P_Y(3)] \\
 &= 1 - \left[\frac{e^{-2} \cdot 2^0}{0!} + \frac{e^{-2} \cdot 2^1}{1!} + \frac{e^{-2} \cdot 2^2}{2!} + \frac{e^{-2} \cdot 2^3}{3!} \right] \\
 &= 1 - e^{-2} \left(1 + 2 + \frac{4}{2} + \frac{8}{6} \right) \\
 &= 1 - e^{-2} \cdot \left(1 + 2 + 2 + \frac{4}{3} \right) \\
 &= 1 - e^{-2} \cdot \frac{19}{3} \\
 &= 1 - \frac{19}{3e^2} \approx 0.1429
 \end{aligned}$$

Special Distributions (PMF)

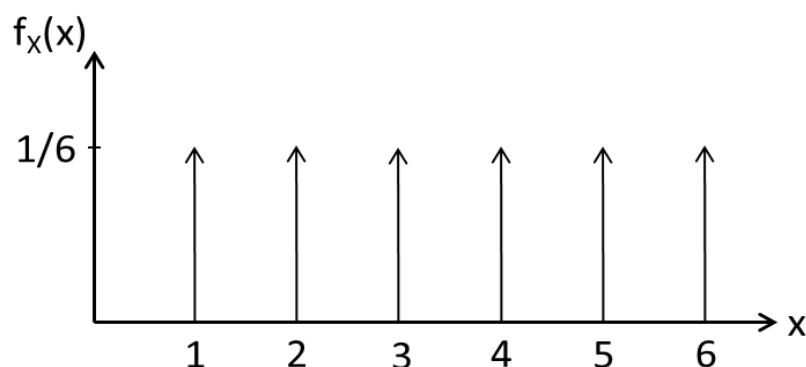
4. Discrete Uniform Distribution

- Models a random experiment where all outcomes in a given integer range are equally likely.
- Example:** Rolling a fair 6-sided die, choosing a random integer from 1 to 10.
- Parameters:**
 - $a, b \in \mathbb{Z}$: lower and upper bounds of the uniform range.
 - $x \in \{a, a + 1, \dots, b\}$: possible outcomes.

- PMF:**

$$P_X(x) = \begin{cases} \frac{1}{b-a+1} & \text{for } x = a, a + 1, \dots, b \\ 0 & \text{otherwise} \end{cases}$$

- Notation:** $X \sim \mathcal{U}[a, b]$



Cumulative Distribution Function (CDF)

The PMF is one way to describe the distribution of a discrete random variable. The **cumulative distribution function** (CDF) of a random variable is another method to describe the distribution of random variables:

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} f_X(x_i)$$

It shows the cumulative probability up to a point.

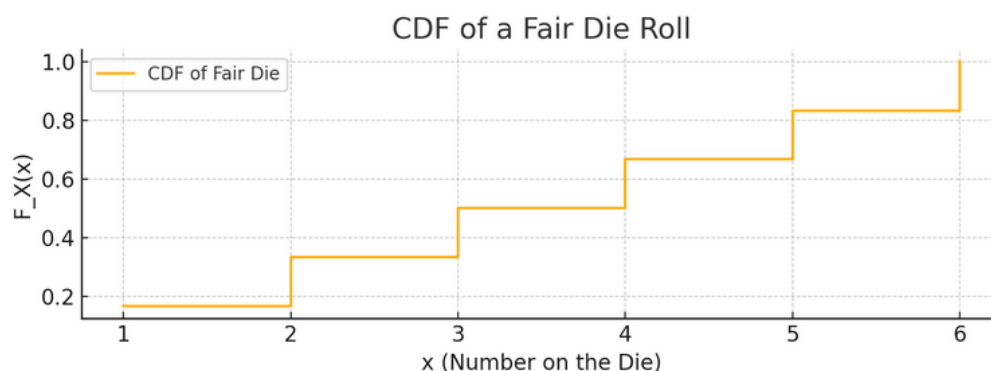
Example: Fair Dice

Let X be the number shown when rolling a fair 6-sided die.

- Each outcome $x = 1, 2, \dots, 6$ has probability $\frac{1}{6}$

Values of the CDF:

$$\begin{aligned} F_X(1) &= P(X = 1) = \frac{1}{6} \\ F_X(2) &= P(X \leq 2) = \frac{2}{6} \\ F_X(3) &= \frac{3}{6}, \quad F_X(4) = \frac{4}{6}, \quad F_X(5) = \frac{5}{6}, \quad F_X(6) = \frac{6}{6} = 1 \end{aligned}$$



Cumulative Distribution Function (CDF)

Example: Sum of Two Fair Dice

Let X be the **sum** of two independent rolls of fair 6-sided dice.

Example calculations:

$$F_X(2) = P(X = 2) = \frac{1}{36}$$

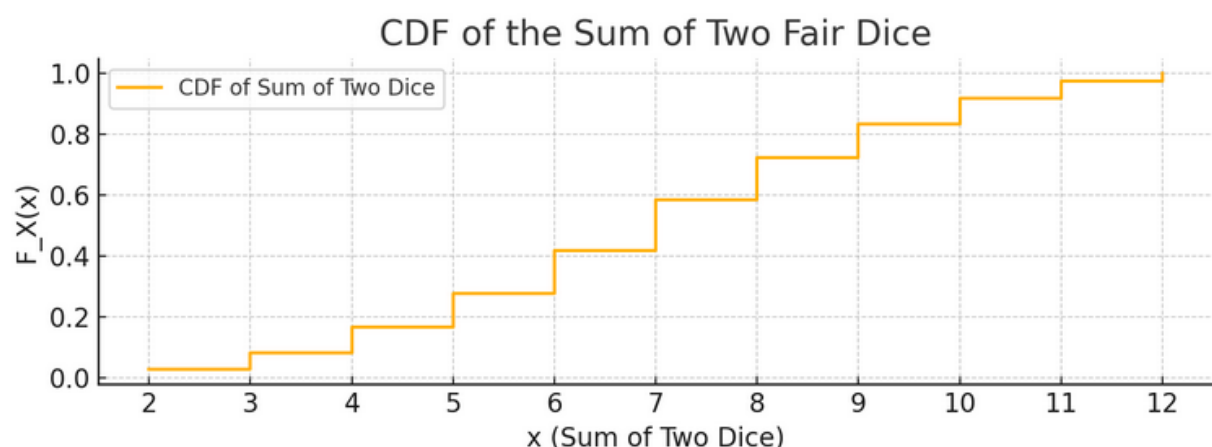
$$F_X(3) = \frac{1 + 2}{36} = \frac{3}{36}$$

$$F_X(4) = \frac{1 + 2 + 3}{36} = \frac{6}{36}$$

$$\vdots$$

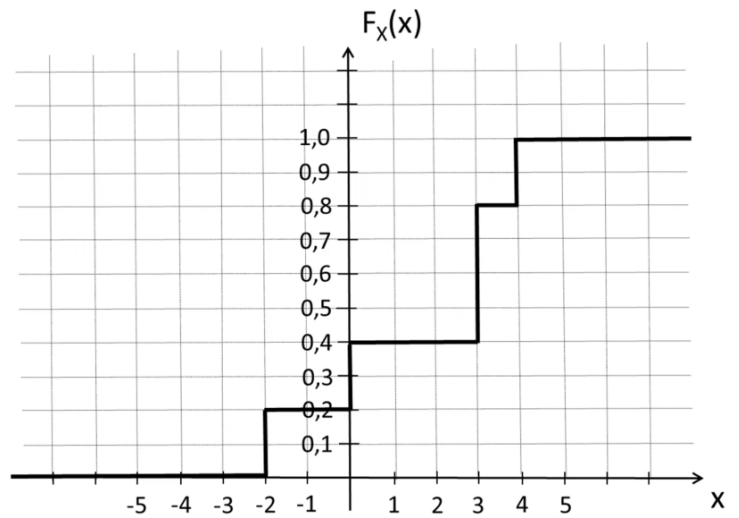
$$F_X(7) = \frac{1 + 2 + 3 + 4 + 5 + 6}{36} = \frac{21}{36}$$

$$F_X(12) = 1$$



Quiz

En diskret stokastisk variabel X har en fordelingsfunktion (cdf) $F_X(x)$ som på figuren.



Hvad er $\Pr(X=3)$?

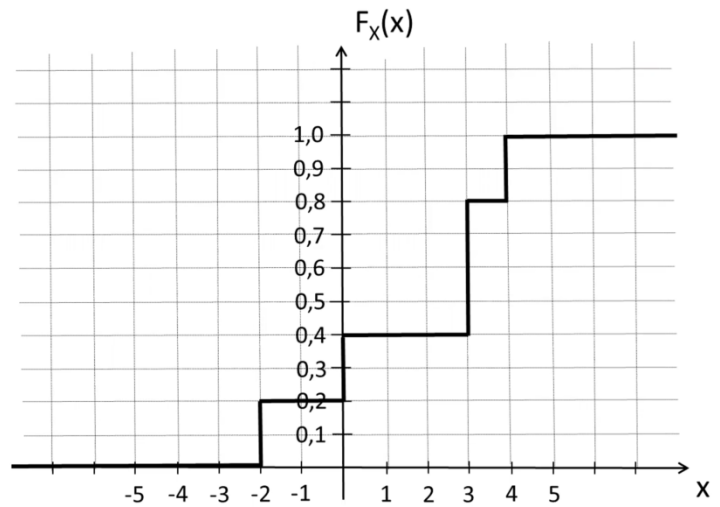
A: 0,8

B: 0,4

C: 0,3

Quiz

En diskret stokastisk variabel X har en fordelingsfunktion (cdf) $F_X(x)$ som på figuren.



Hvad er $\Pr(X=1)$?

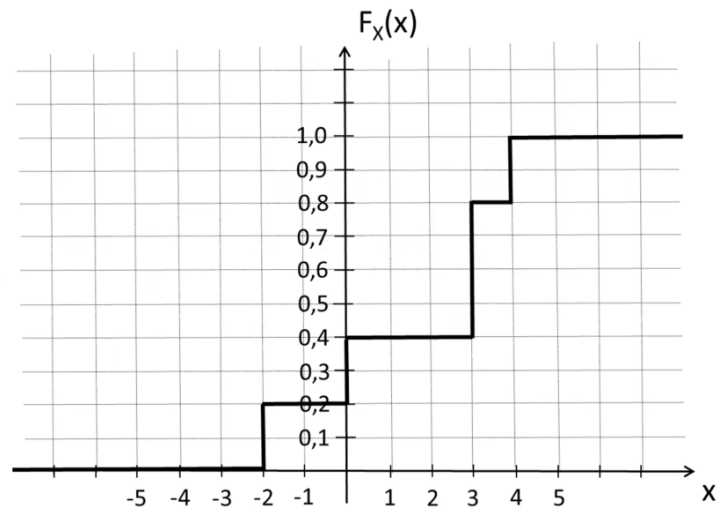
A: 0

B: 0,2

C: 0,4

Quiz

En diskret stokastisk variabel X har en fordelingsfunktion (cdf) $F_X(x)$ som på figuren.



Hvad er $\Pr(-1 \leq X < 4)$? A: 0,2

B: 0,6

C: 0,8

Sample mean and Expectation

Example: Average vs expectation value

Let's say you toss a fair die

N = 3 Rolls

- Sample values: [4, 5, 3]
- Average calculation:

$$\frac{4 + 5 + 3}{3} = \frac{12}{3} = 4.0$$

N = 10 Rolls

- Sample values: [5, 5, 2, 3, 3, 3, 5, 4, 3, 6]
- Average calculation:

$$\frac{5 + 5 + 2 + 3 + 3 + 3 + 5 + 4 + 3 + 6}{10} = \frac{39}{10} = 3.9$$

N = 20 Rolls

- Sample values:
[4, 6, 1, 2, 3, 5, 6, 3, 4, 2, 5, 6, 2, 1, 4, 3, 5, 2, 4, 3]

- Average calculation:

$$\frac{4 + 6 + 1 + 2 + 3 + 5 + 6 + 3 + 4 + 2 + 5 + 6 + 2 + 1 + 4 + 3 + 5 + 2 + 4 + 3}{20} = \frac{71}{20} = 3.55$$

If we roll a die many times and keep track of how often each number comes up, the **sample mean** value of all results will tend to get closer to the **true mean denoted $E[X]$** :

As you repeat the experiment more and more, the sample average gets closer to the expectation:

$$\text{Average of observations} \longrightarrow \mathbb{E}[X] \quad \text{as } N \rightarrow \infty$$

Law of large numbers (LLN)

Example: Rolling a fair die

Event: $E = \{5\}$, so the theoretical probability is

$$P(E) = \frac{|E|}{|S|} = \frac{1}{6} \approx 0.167.$$

- Estimate from data:

- After $N = 10$ rolls

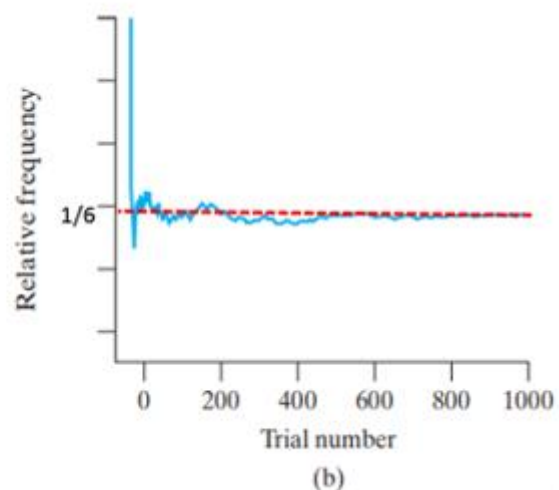
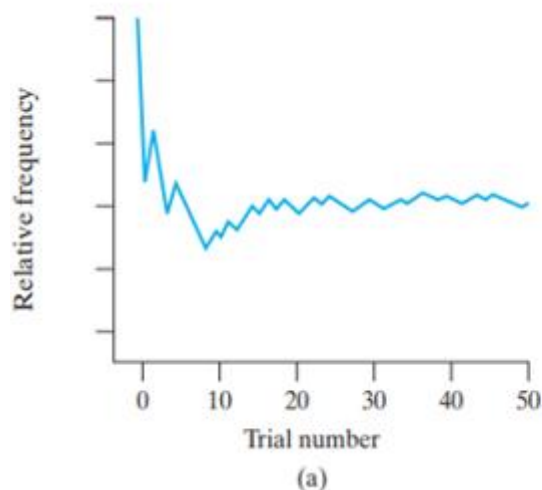
$$\frac{N_5}{10} \approx 0.2$$

- After $N = 50$ rolls

$$\frac{N_5}{50} \approx 0.18$$

- After $N = 500$ rolls

$$\frac{N_5}{500} \approx 0.166$$



Law of large numbers: As N increases, $\frac{N_5}{N} \rightarrow P(E)$

Expectation

Suppose we observe a discrete random variable X over N trials.
Let N_k be the number of times the value x_k occurs.

Sample mean:

$$\bar{X}_N = \frac{x_1 N_1 + x_2 N_2 + \cdots}{N} = \sum_k x_k \frac{N_k}{N}.$$

Here $\frac{N_k}{N}$ are empirical frequencies.

By the Law of Large Numbers,

$$\frac{N_k}{N} \rightarrow P(X = x_k), \quad N \rightarrow \infty,$$

so

$$\bar{X}_N \rightarrow \sum_k x_k P(X = x_k) = E[X].$$

Apply formula to the die:

$$E[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{21}{6} = 3.5$$

Expectation

Example: expectation value of 5 coin flips



Let X be the number of heads in 5 flips of a fair coin.

The possible values of X are 0, 1, 2, 3, 4, 5.

Since $X \sim \text{Binomial}(n = 5, p = 0.5)$, the probability mass function is

$$P(X = k) = \binom{5}{k} (0.5)^k (0.5)^{5-k} = \binom{5}{k} (0.5)^5.$$

The expected value is computed as

$$E[X] = \sum_{k=0}^5 k \cdot P(X = k) = \sum_{k=0}^5 k \cdot \binom{5}{k} (0.5)^5 = 2.5.$$

Mathcad calculation:

$$E := \sum_{k=0}^5 k \cdot \text{combin}(5, k) \cdot 0.5^5 = 2.5$$

Expectation

The coin flip example shows that the expectation calculations can be complicated.

So we derive **specialized expectation formulas** for each distribution

Expectation value formula:

1. Bernoulli Distribution

Let $X \sim \text{Bernoulli}(p)$. That is, $X = 1$ with probability p , and $X = 0$ with probability $1 - p$.

$$\mathbb{E}[X] = 0 \cdot (1 - p) + 1 \cdot p = p$$

Expectation

Before showing the expectation formula for the Binomial distribution, we need an **important result**:

Theorem: Linearity of Expectation

- For any constants $a, b \in \mathbb{R}$ and random variable X :

$$\mathbb{E}[aX + b] = a \cdot \mathbb{E}[X] + b$$

- For any finite collection of random variables $X_1, X_2, \dots, X_n$:

$$\mathbb{E}[X_1 + X_2 + \dots + X_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \dots + \mathbb{E}[X_n]$$

Expectation

Expectation value formula:

2. Binomial Distribution

Let $X \sim \text{Binomial}(n, p)$. That is, X is the number of successes in n independent Bernoulli(p) trials.

So we can write:

$$X = X_1 + X_2 + \cdots + X_n$$

where each $X_i \sim \text{Bernoulli}(p)$, meaning:

$$\mathbb{E}[X_i] = 0 \cdot (1 - p) + 1 \cdot p = p$$

Apply Linearity of Expectation

Using the rule:

$$\mathbb{E}[X_1 + X_2 + \cdots + X_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n]$$

We get:

$$\mathbb{E}[X] = \mathbb{E}[X_1 + X_2 + \cdots + X_n] = p + p + \cdots + p = np$$

According to this formula the expectation value of 5 coin flips is:

$$E[X] = 5 \cdot 0.5 = 2.5$$

This is in agreement with our previous result!

Expectation

3. Poisson Distribution

Let $X \sim \text{Poisson}(\lambda)$.

That is, X is the number of events occurring in a fixed time interval, and:

- $\lambda > 0$ is the expected **number of events** in that interval.

We use the formula:

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k \cdot P(X = k) = \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda} \lambda^k}{k!}$$

After simplification (see details in the book), the result is:

$$\mathbb{E}[X] = \lambda$$

4. Discrete Uniform Distribution

Let $X \sim \mathcal{U}(a, b)$.

That is, X is uniformly distributed on the integers from a to b .

So each outcome is equally likely.

We use the formula:

$$\mathbb{E}[X] = \frac{a + b}{2}$$

Expected Value Of A Functions Of Random Variables

Example: (1) Calculate PMF of $Y = 2|X|$

Let:

$$P_X(k) = \frac{1}{5}, \quad \text{for } k \in \{-1, 0, 1, 2, 3\}$$

The range (support) of X is

$$R_X = \{-1, 0, 1, 2, 3\}.$$

And:

$$R_Y = \{0, 2, 4, 6\}$$

Using the definition of PMF:

$$P_Y(0) = P(X = 0) = \frac{1}{5}$$

$$P_Y(2) = P(X = -1 \text{ or } X = 1) = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}$$

$$P_Y(4) = P(X = 2) = \frac{1}{5}$$

$$P_Y(6) = P(X = 3) = \frac{1}{5}$$

So the PMF of Y is:

$$P_Y(k) = \begin{cases} \frac{1}{5} & \text{for } k = 0, 4, 6 \\ \frac{2}{5} & \text{for } k = 2 \\ 0 & \text{otherwise} \end{cases}$$

(2) Calculate **expectation value of $Y = 2|X|$**

$$\mathbb{E}[Y] = \sum_{y \in R_Y} y \cdot P_Y(y)$$

$$= 0 \cdot \frac{1}{5} + 2 \cdot \frac{2}{5} + 4 \cdot \frac{1}{5} + 6 \cdot \frac{1}{5} = \frac{0 + 4 + 4 + 6}{5} = \frac{14}{5} = 2.8$$

Expected Value Of A Functions Of Random Variables

If X is a random variable and $Y = g(X)$, then Y is also a random variable. **PMF of Y** is given by:

$$P(g(X) = y) = \sum_{x: g(x)=y} P_X(x)$$

There is another way which is sometimes easier. It is called the law of the unconscious statistician (**LOTUS**).

$$\mathbb{E}[g(X)] = \sum_{x_k \in R_X} g(x_k) \cdot P_X(x_k)$$

This skips the step of finding the distribution of $Y = g(X)$.

Example: Given $Y = 2|X|$, calculate $E[Y]$ with **LOTUS**

$$\begin{aligned} \mathbb{E}[Y] &= \sum_x 2|x| \cdot \frac{1}{5} = 2(|-1| + |0| + |1| + |2| + |3|) \cdot \frac{1}{5} \\ &= 2 \cdot (1 + 0 + 1 + 2 + 3) \cdot \frac{1}{5} = \frac{14}{5} = 2.8 \end{aligned}$$

Variance

Definition of Variance

The **variance** measures how spread out a distribution is around the mean.

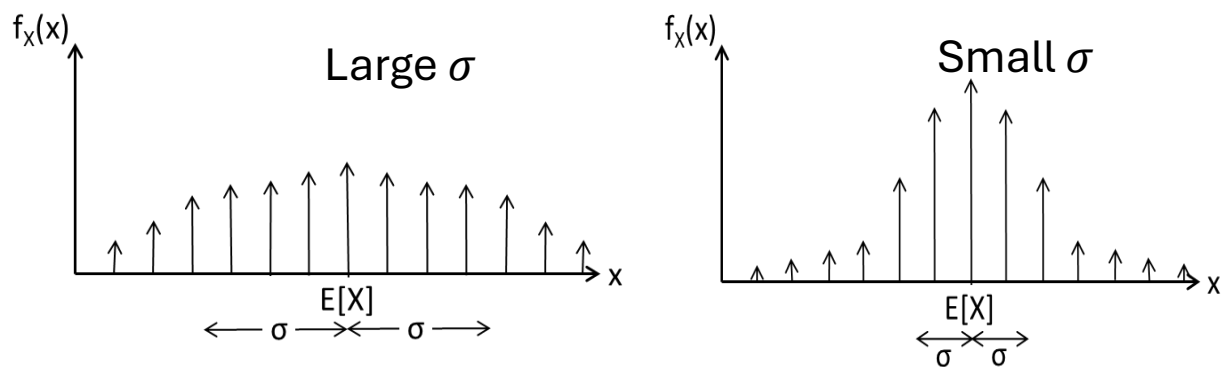
$$\text{Var}(X) = \mathbb{E}[(X - \mu_X)^2]$$

- Always non-negative.
- Large when values of X are far from μ_X .

$$\mu_X = \mathbb{E}[X]$$

Standard Deviation

$$\text{SD}(X) = \sigma_X = \sqrt{\text{Var}(X)}$$



Variance

Computing Variance Using LOTUS

We use the Law of the Unconscious Statistician (LOTUS) with $g(X) = (X - \mu_X)^2$:

$$\text{Var}(X) = \sum_{x_k \in \mathcal{R}_X} (x_k - \mu_X)^2 \cdot P_X(x_k)$$

Example:

Two random variables can have the **same mean** but very different **spread**.

Example:

$$P_X(x) = \begin{cases} 0.5 & \text{if } x = -100 \text{ or } x = 100 \\ 0 & \text{otherwise} \end{cases} \quad P_Y(y) = \begin{cases} 1 & \text{if } y = 0 \\ 0 & \text{otherwise} \end{cases}$$

Both have mean 0:

$$\mathbb{E}[X] = \mathbb{E}[Y] = 0$$

But X varies a lot from its mean, while Y is always 0.

$$\text{Var}(X) = (-100)^2 \cdot 0.5 + (100)^2 \cdot 0.5 = 10000$$

$$\text{Var}(Y) = 0^2 \cdot 1 = 0$$

Variance

We can derive a **practical formula** for the variance:

$$\text{Var}(X) = \mathbb{E}[(X - \mu_X)^2]$$

Expand the square

$$(X - \mu_X)^2 = X^2 - 2\mu_X X + \mu_X^2$$

Apply expectation (linearity of expectation)

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2 - 2\mu_X X + \mu_X^2] \\ &= \mathbb{E}[X^2] - 2\mu_X \mathbb{E}[X] + \mathbb{E}[\mu_X^2] \\ &= \mathbb{E}[X^2] - 2\mu_X^2 + \mu_X^2 = \mathbb{E}[X^2] - \mu_X^2\end{aligned}$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Variance

Example: Fair Die

Let $X \in \{1, 2, 3, 4, 5, 6\}$ with equal probabilities.

$$\mathbb{E}[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{7}{2}$$

$$\begin{aligned} \mathbb{E}[X^2] &= \sum_{k=1}^6 k^2 \cdot \frac{1}{6} = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} \\ &= \frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6} \end{aligned}$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{49}{4} \approx 2.92$$

$$\sigma_X = \sqrt{2.92} \approx 1.71$$

Interpret the result

Here, $\mu \pm \sigma = 3.5 \pm 1.71 \approx [1.8, 5.2]$.

That interval indeed covers the outcomes $\{2, 3, 4, 5\}$, which is 4 out of 6 possible results ($\approx 67\%$).

Variance

Properties of Variance

1: Not a Linear Operator

$$\text{Var}(aX + b) = a^2 \cdot \text{Var}(X)$$

Scaling changes variance; shifting does not.

Variance NOT a linear operation!

2: Additive Over Independent RVs

If $X = X_1 + \dots + X_n$ and X_i are independent:

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i)$$

Example

Let $X \sim \text{Binomial}(n, p)$. We want to find $\text{Var}(X)$.

A Binomial variable counts the number of successes in n independent Bernoulli trials:

$$X = X_1 + X_2 + \dots + X_n, \quad \text{with } X_i \sim \text{Bernoulli}(p)$$

So:

$$\text{Var}(X_i) = \mathbb{E}[X_i^2] - (\mathbb{E}[X_i])^2 = p - p^2 = p(1 - p)$$

Since the X_i are **independent**: $\text{Var}(X) = \sum \text{Var}(X_i)$

Final Step:

$$\text{Var}(X) = \sum_{i=1}^n p(1 - p) = np(1 - p)$$

Quiz

Two independent random variables X and Y are given with:

$$\mathbb{E}[X] = 3, \quad \text{Var}(X) = 1, \quad \mathbb{E}[Y] = 4, \quad \text{Var}(Y) = 3$$

A new random variable Z is defined as:

$$Z = 3X - 2Y - 5$$

What is the expected value $\mathbb{E}[Z]$?

A: -4 B: 1 C: 12

Quiz

Two independent random variables X and Y are given with:

$$\mathbb{E}[X] = 3, \quad \text{Var}(X) = 1, \quad \mathbb{E}[Y] = 4, \quad \text{Var}(Y) = 3$$

A new random variable Z is defined as:

$$Z = 3X - 2Y - 5$$

What is the variance $\text{Var}(Z)$?

A: -8 B: 16 C: 21

Engelsk – dansk oversættelse

English Term	Danish Translation
Random variable	Stokastisk variabel / Tilfældig variabel
Range (of a random variable)	Værdimængde (af en stokastisk variabel)
Probability Mass Function (PMF)	Sandsynlighedsmassefunktion (PMF)
Cumulative Distribution Function (CDF)	Kumulativ fordelingsfunktion (CDF)
Expectation / Expected value	Forventningsværdi / Middelværdi
Linearity of Expectation	Lineæritet af forventningsværdi
Law of the Unconscious Statistician (LOTUS)	Statistikers ubevidste lov (LOTUS)
Function of a random variable	Funktion af en stokastisk variabel
Variance	Varians
Mean	Gennemsnit
Distribution	Fordeling
Uniform distribution	Uniform fordeling
Poisson distribution	Poisson-fordeling